



Realize Your Career Dreams

GIFT School of Engineering and Applied Sciences

Spring 2019

CS-240: Object-oriented Programming

Lab-6 Manual

Basics of Classes and Objects

Task #1: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Write a class definition **DayOfYear** with two data member **day**, and **month** of integer type. It also contains the methods **input**, **output**, **set**, **getMonth**, and **getDay**. These methods should perform the actions as described in the **procedure** below:

Procedure

- Build a class with two **private** data member **day** and **month** of integer type for holding data like birthdays and holiday.
- This class also contains methods described as follows:

- **void input()**

Gives input to the data members of the object. Use a **Scanner** object to read input. Here's an example:

```
Scanner read = new Scanner(System.in);
```

```
DayOfYear obj = new DayOfYear();
```

```
System.out.println("Enter the month as a number: ");  
int m = read.nextInt();
```

```
System.out.println("Enter the day of the month: ");  
int d = read.nextInt();
```

```
//call the set method and pass values  
set(m, d);
```

- **void output()**

Displays the month and day of the year.

- **void set(int m, int d)**

Assign new values to the data members of the object.

- **int getMonth()**

Returns the month

- **int getDay()**

Returns the day of the month

- Create two objects **today** and **birthday** of **DayOfYear** in the **main()** method.
- Call the **input** method of **today** object for input from the user in today's data members. Display these on the screen.

- Assign values to **birthday** members using **set** method and display these on the screen.

Exercise Output

```
Enter today's date :  
  
Enter the month as a number : 3  
Enter the day of the month : 21  
Today's date is month = 3, day = 21  
  
Ali's birthday is month = 3, day = 21  
Happy Birthday Ali!
```

1. Create a program called **DayOfYear.java** for the class, and **RunDayOfYear.java** having the **main** method.
2. Correctly display appropriate messages.

Task #2: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Write a class definition **Book** with three data member **bookId (int)**, **pages (int)** and **price (double)**. The class has the following methods as described in the **procedure** below:

Procedure

- Build a class with three **private** data member **bookId**, **pages** and **price** for holding data about books.
- This class also contains the following methods:
 - **void get()**
Gives input to the data members of the object. Use a **Scanner** object for getting the input. You will then pass these values to the **set** method.
 - **void show()**
Displays all the values of data members of an object.
 - **void set(int b, int p, double pr)**
Assigns new values to the data members of the object.
 - **Setter** and **Getter** methods
Create all setter and getters methods
- Create two objects **maths** and **english** in the **main** method.
- Call **get** function of **maths** for input from the user in **maths** data members. Assign value to this object using the **set** method.
- Do the same for the **english** object and use the **set** method to assign values.
- Display the values of **maths** and **english** objects.
- Change values of the **maths** object by using its *setter* functions.
- Display the values of the **maths** object by using the *getter* functions.
- Show the details of the costliest book amongst **maths** and **english**.

Exercise Output

```
Enter Book ID: 1
Enter Pages: 250
Enter Price: 125.20

The details of the costliest book are:
BookID = 2
Pages = 320
Price = 150.75
```

1. Create a program called **Book.java** for the class, and **RunBook.java** having the **main** method.
2. Correctly display appropriate messages.

Task #3: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Design a class named **Pet**, which should have the following fields:

- **name**. The name field holds the name of a pet.
- **animal**. The animal field holds the type of animal that a pet is. Example values are "Dog", "Cat", and "Bird".
- **age**. The age field holds the pet's age.

The **Pet** class should also have the following methods:

- **setName**. The **setName** method stores a value in the **name** field.
- **setAnimal**. The **setAnimal** method stores a value in the **animal** field.
- **setAge**. The **setAge** method stores a value in the **age** field.
- **getName**. The **getName** method returns the value of the **name** field.
- **getAnimal**. The **getAnimal** method returns the value of the **animal** field.
- **getAge**. The **getAge** method returns the value of the **age** field.

1. Create a program called **Pet.java** for the class, and **RunPet.java** having the **main** method.
2. Create objects of all types of pets, such as a **Dog**, **Cat**, and a **Bird**.
3. Set the object's values using the **set** methods, and display the values using the **get** methods.

Task #4: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Look at the following partial class definition, and then respond to the questions that follow it:

```
class BookDescription {  
    private String title;  
    private String author;  
    private String publisher;  
    private int copiesSold;  
} //class
```

- Write accessor (**get**) and mutator (**set**) methods for each field.
- Write a **display** method to print all values of the object.
- Create at least **three** objects of the class having different values.
- Call all setter and getter methods on each object. Call the **display** method for all objects.

1. Create a program called **BookDescription.java** for the class, and **RunBookDescription.java** having the **main** method.
2. Correctly display appropriate messages.

Task #5: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Write a class named **Employee** that has the following fields:

- **name**. The **name** field references a **String** object that holds the employee's name.
- **idNumber**. The **idNumber** is an **int** variable that holds the employee's ID number.
- **department**. The **department** field references a **String** object that holds the name of the department where the employee works.
- **position**. The **position** field references a **String** object that holds the employee's job title.

Write appropriate mutator methods that store values in these fields and accessor methods that return the values in these fields.

Once you have written the class, write a separate program that creates three **Employee** objects to hold the following data:

Name ID	Number	Department	Position
M Ali	47899	Accounting	Vice President
Baber Jameel	39119	IT	Programmer
Rida Naeem	81774	Manufacturing	Engineer

The program should store this data in the three objects and then display the data for each employee on the screen.

1. Create a program called **Employee.java** for the class, and **RunEmployee.java** having the **main** method.
2. Correctly display appropriate messages.

Task #6: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Write a class named **Car** that has the following fields:

- **yearModel**. The **yearModel** field is an **int** that holds the car's year model.
- **make**. The **make** field references a **String** object that holds the make of the car.
- **speed**. The **speed** field is an **int** that holds the car's current speed.

In addition, the class should have the following methods:

- **Accessors**. Appropriate accessor methods should get the values stored in an object's **yearModel**, **make**, and **speed** fields.
- **accelerate**. The **accelerate** method should add **5** to the **speed** field each time it is called.
- **brake**. The **brake** method should subtract **5** from the **speed** field each time it is called.

Demonstrate the class in a program that creates a **Car** object, and then calls the **accelerate** method five times. After each call to the **accelerate** method, get the current **speed** of the car and display it. Then call the **brake** method five times. After each call to the **brake** method, get the current **speed** of the car and display it.

1. Create a program called **Car.java**, and **RunCar.java** having the **main** method.
2. Correctly display appropriate messages.

Task #7: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Write a class named **RetailItem** that holds data about an item in a retail store. The class should have the following fields:

- **description**. The **description** field references a **String** object that holds a brief description of the item.
- **unitsOnHand**. The **unitsOnHand** field is an **int** variable that holds the number of units currently in inventory.
- **price**. The **price** field is a **double** that holds the item's retail price.

Write appropriate mutator methods that store values in these fields, and accessor methods that return the values in these fields.

Once you have written the class, write a separate program that creates three **RetailItem** objects and stores and then displays the following data in them:

	Description	Units on Hand	Price
Item #1	Jacket	12	59.95
Item #2	Designer Jeans	40	34.95
Item #3	Shirt	20	24.95

1. Create a program called **RetailItem.java**, and **RunRetailItem.java** having the **main** method.
2. Correctly display appropriate messages.

Task #8: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Design a **Payroll** class that has fields for an employee's **name (String)**, **idNumber (int)**, **hourlyPayRate (double)**, and **numberOfHoursWorked (double)**.

Write the appropriate accessor and mutator methods. The class should also have a method that returns the employee's **gross pay (grossPay)**, which is calculated as the number of hours worked multiplied by the hourly pay rate.

Write a program that demonstrates the class by creating three different **Payroll** objects, then asking the user to enter the data for each employee. The program should display the amount of gross pay earned by each employee.

1. Create a program called **Payroll.java**, and **RunPayroll.java** having the **main** method.
2. Correctly display appropriate messages.

Task #9: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

Write a **Temperature** class that will hold a temperature in Fahrenheit, and provide methods to get the temperature in Fahrenheit, Celsius, and Kelvin. The class should have the following field:

- **ftemp** – A **double** that holds a Fahrenheit temperature.

The class should have the following methods:

- **setFahrenheit** – The **setFahrenheit** method accepts a Fahrenheit temperature (as a **double**) and stores it in the **ftemp** field.
- **getFahrenheit** – Returns the value of the **ftemp** field, as a Fahrenheit temperature (no conversion required).
- **getCelsius** – Returns the value of the **ftemp** field converted to Celsius.
- **getKelvin** – Returns the value of the **ftemp** field converted to Kelvin.

Use the following formula to convert the Fahrenheit temperature to Celsius:

$$\text{Celsius} = (5/9) \times (\text{Fahrenheit} - 32)$$

Use the following formula to convert the Fahrenheit temperature to Kelvin:

$$\text{Kelvin} = ((5/9) \times (\text{Fahrenheit} - 32)) + 273$$

Demonstrate the **Temperature** class by writing a separate program that asks the user for a Fahrenheit temperature. The program should create an instance of the **Temperature** class, ask the user for an input Fahrenheit temperature, and call the **setFahrenheit** method to set the data member value. The program should then call the object's methods to display the temperature in Celsius and Kelvin.

1. Create a program called **Temperature.java**, and **RunTemperature.java** having the **main** method.
2. Correctly display appropriate messages.

Task #10: Creating Classes and Objects

In this task, you are being asked to write a class and create objects in Java.

NOTE: Write your class and the *main* method in separate files.

The following table lists the freezing and boiling points of several substances.

Substance	Freezing Point	Boiling Point
Ethyl Alcohol	−173	172
Oxygen	−362	−306
Water	32	212

Design a class called **Substance** that stores a temperature in a **temperature (int)** field and has the appropriate accessor and mutator methods for the field. The class should also have the following methods:

- **isEthylFreezing**. This method should return the **boolean** value **true** if the temperature stored in the **temperature** field is at or below the freezing point of ethyl alcohol. Otherwise, the method should return **false**.
- **isEthylBoiling**. This method should return the **boolean** value **true** if the temperature stored in the **temperature** field is at or above the boiling point of ethyl alcohol. Otherwise, the method should return **false**.
- **isOxygenFreezing**. This method should return the **boolean** value **true** if the temperature stored in the **temperature** field is at or below the freezing point of oxygen. Otherwise, the method should return **false**.
- **isOxygenBoiling**. This method should return the **boolean** value **true** if the temperature stored in the **temperature** field is at or above the boiling point of oxygen. Otherwise, the method should return **false**.
- **isWaterFreezing**. This method should return the **boolean** value **true** if the temperature stored in the **temperature** field is at or below the freezing point of water. Otherwise, the method should return **false**.
- **isWaterBoiling**. This method should return the **boolean** value **true** if the temperature stored in the **temperature** field is at or above the boiling point of water. Otherwise, the method should return **false**.

Write a program that demonstrates the class. The program should ask the user to enter a temperature, and then display a list of the substances that will freeze at that temperature and those that will boil at that temperature. For example, if the temperature is **-20** the class should report that **water** will **freeze**, and **oxygen** will **boil** at that temperature.

1. Create a program called **Substance.java**, and **RunSubstance.java** having the **main** method.
2. Correctly display appropriate messages.